

1 **Tracing the carbon footprint of cotton garments based on their life cycle: Evidence from**
2 **an empirical study of multiple sites in China**

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21 **ABSTRACT**

22 Greenhouse gas emissions from Chinese textile production are critical contributors to global

climate change that vary greatly based on the type of textile. Environmental impacts of five clothing types in China were analyzed based on a cross-provincial survey and life cycle assessment. Carbon footprints of different clothing showed significant variations: childrenswear (2.20 kg CO₂eq), knitwear (6.58 kg CO₂eq), fatigues (37.70 kg CO₂eq), T-shirts (8.88 kg CO₂eq), and workwear (18.68 kg CO₂eq). In which key driver factors include agricultural film use (27.57%), electricity consumption (38.15%), and fertilizer application (17.77%). Textile production impacts depend on energy sources, processing techniques, and factory size. Four scenarios—utilizing clean energy (57.58%), optimizing factory sizes (25.19%–30.43%), organic cultivation (64.2%), and recycling (21.75%–58.85%)—suggest ways to mitigate environmental effects and support sustainable apparel development, providing insights for global cotton industry emission reduction strategies.

Keywords: Carbon footprint; cotton textile; sustainable production; environmental impact; life cycle assessment (LCA)

1. Introduction

Global warming intensifies, as indicated by the Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report on Climate Change, which notes a 1.1°C increase in global surface temperatures for the period from 2011 to 2020 compared to those recorded between 1850 and 1900 (IPCC, 2021). Textiles contribute 5–10% of global greenhouse gas (GHG) emissions (Zhang et al., 2022), and their life cycle carbon emissions are second to those of petroleum and natural gas and exceed those of international aviation and maritime transport systems combined (European Parliament, 2021). From 2022 to 2023, cotton, which is the largest fibre crop, experienced its global production predominantly led by China, India, and the

United States, which together accounted for more than half of the total yield (Shahbandeh, 2023). As one of the largest cotton spinning countries, China accounted for 32% of the total cotton used worldwide (Meyer and Dew, 2023). The textile industry in China significantly contributes to the global market, with textile companies from this country contributing nearly 30% of textiles worldwide as of 2010 (Astrid Pepermans, 2019; Wang et al., 2015). The textile and apparel sector in China releases approximately 230 million tonnes of emissions annually, highlighting a significant area and need for emission reductions (China daily, 2023). While cotton textiles are crucial, they inevitably lead to significant environmental impacts. Therefore, it is crucial to measure the environmental impact of products from the cotton textile sector in China to advance strategies for emission reductions throughout the textile value chain.

Life cycle assessment (LCA) is a pivotal tool for quantifying environmental impacts that has been widely applied to various major textile products (Baydar et al., 2015). However, there are few comprehensive seed-to-garment analyses focusing on specific clothing types. Generally, the life cycle of cotton begins with planting and continues to light industrial production after harvest before undergoing ginning, spinning, knitting, dyeing, and making-up. Specifically, cotton seeds undergo a series of complex processes, including the stages of germination, seedling, boll, and mature harvesting. The planting phase involves a variety of inputs, often leading to relatively significant environmental impacts (Islam et al., 2024). For instance, Günther compared the carbon footprint (CFP) of cotton produced in Xinjiang with that of jute, a substitute fibre, and found that reducing the use of chemical fertilizers can decrease the CFP of cotton (Günther et al., 2017). The processing phase—the primary contributor to carbon emissions in the cotton textile industry—is further subdivided into raw cotton decontamination,

yarn production, and fabric making-up stages (Subramaniam et al., 2024). During the initial phase of converting cotton into yarn, the first task involves purifying the material by reducing its impurity content from an initial value of 16% to a value between 1.8% and 2.5% (Zhang et al., 2015). Yarn production advances through spinning, which transforms fibres into yarn. This transformation is followed by converting yarn into fabric, a process differentiated by either weaving or knitting, contingent on the technique applied. Notably, short fibres are associated with a smaller CFP than long fibres (Bianco et al., 2023).

Dyeing is a critical process in the textile industry, embodying a multifaceted procedure that necessitates a variety of chemical inputs. This procedure involves several pivotal stages, including pretreatment, bleaching, dyeing, soaping, dewatering, drying, and, occasionally, printing (Seo et al., 2024). Notably, in certain instances, yarn might undergo dyeing prior to the knitting process (Raza et al., 2024). Research indicates that colour does not markedly influence the CFP in the industrial fabrication of textiles (Yan et al., 2014). For instance, Wang et al. (2023) reported that in the dyeing and finishing stages of polyester–cotton knitted fabrics, steam consumption is the predominant contributor to the CFP. The garment making-up process, which encompasses cutting, manufacturing, ironing, and packaging, predominantly relies on electricity as the principal input at this life cycle stage (Barani and Montazer, 2008).

The CFPs of cotton textiles vary notably across different countries. For instance, in the production of cotton shirts in China, the making-up stage accounts for the minimal CFP (Wang et al., 2015). Conversely, the making-up stage of jeans production in Brazil has the greatest environmental impact (Morita et al., 2020). These disparities largely stem from variations in the production techniques, scales, patterns, and types of energy utilized. To mitigate carbon

emissions in the cotton life cycle, scholars have recently advocated for reducing energy consumption via technological advancements (Fonseca et al., 2023), transitioning from coal to renewable energy sources (Chen et al., 2021), and shortening global supply chains through nearby or redistributed production to decrease logistics-associated emissions (Leal Filho et al., 2022). Furthermore, textile recycling has emerged as a vital approach for diminishing environmental impacts throughout the production process (Zhang et al., 2022). These measures facilitate the transition of the textile industry towards sustainable development.

Despite the extensive body of research on the apparel industry chain, a significant portion of these investigations are confined to isolated points of study or to merely synthesizing literature data, leaving a substantial gap in knowledge concerning investigations of the entire industry chain through empirical research. The variations in economic, environmental, and social policies across different regions render the comparability of studies conducted in different locations challenging; moreover, literature analyses are limited to re-evaluating past findings without examining the underlying mechanisms responsible for these outcomes (Niinimäki et al., 2020). To bridge these research gaps, multipoint field research data are leveraged from China. An LCA-based CFP evaluation method is utilized to separately calculate the CFPs of various types of garments. This approach elucidates the breakdown of CFPs across different stages for various garments. Furthermore, this technique compares the environmental impacts across the life cycle stages of different garment types using LCA, ultimately quantifying the emission reduction potentials of four distinct scenarios. The findings offer a practical reference for formulating environmental policies and emission reduction strategies pertinent to cotton cultivation and to the sustainable production and consumption of garments.

2. Methodology

2.1. Case study and functional unit

The study was derived from field research conducted between April and November 2023 across Henan (36°06'N, 114°35'E) and Xinjiang (39°93'N, 116°59'E) Provinces. Herein, five categories of cotton garments wholly manufactured in China—childrenswear, knitwear, fatigues, T-shirt, and workwear—were analysed. The investigation examined principal production subprocesses and the utilization of different types of energy, materials, and machinery. In this study, the average weights of five types of clothing are shown in Table A.1, with each type of garment and per unit weight garment serving as a functional unit under investigation. The calculations account for production-related quality degradation, including standard material waste and quality deficits.

2.2. Assessment framework and system boundary

To accurately assess GHG emissions within the textile industry, establishing a definitive boundary is essential. The boundary definitions in our study were based on the Greenhouse Gas Protocol Corporate Accounting and Reporting Standard (Bhatia et al., 2004).

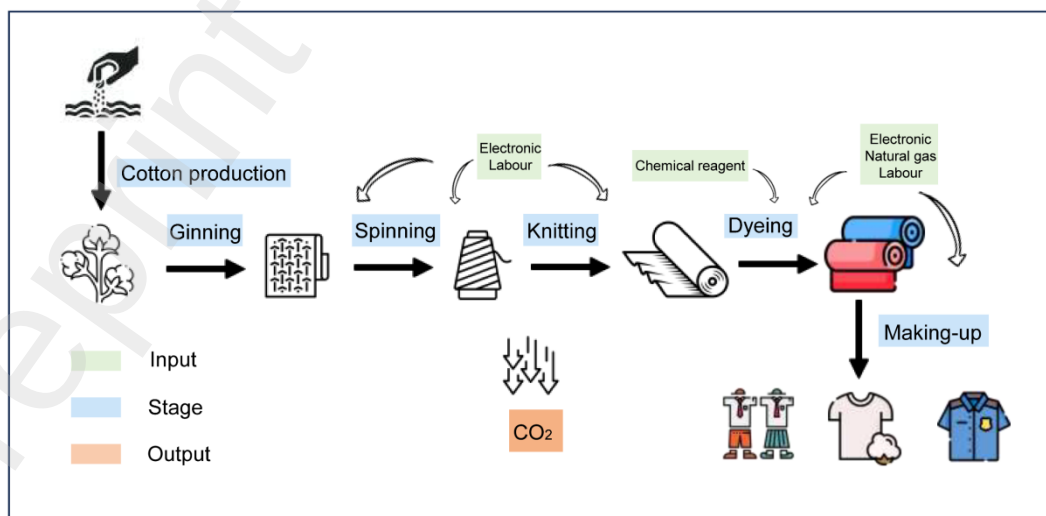


Fig. 1. Research system boundary and processes of the whole cotton industry chain. Single hollow arrows indicate inputs, the items with a blue background indicate each stage, and the items with a green background represent the input categories.

This research delineated a seed-to-garment boundary that featured six pivotal stages: cotton cultivation, raw cotton production, yarn spinning, fabric creation, fabric dyeing, and garment manufacturing (Fig. 1). The CFP associated with cotton production encompassed both direct GHGs released during the cultivation of cotton and indirect GHGs generated from the use of agricultural inputs (Huang et al., 2022). Direct GHG emissions from cotton cultivation included diesel combustion and nitrous oxide release, while indirect emissions stemmed from the electricity used in irrigation and the application of fertilizers, agricultural films, and pesticides. The cotton-producing region of Xinjiang accounted for 90% of the total cotton output in China. In this study, the CFP compositions of Xinjiang cotton from 2017 to 2021 were considered, with the results representing the five-year average. Our comprehensive CFP evaluation considered every input and output across each subprocess. As outlined in Table A.2 and Table A.3, the ginning, spinning, and knitting phases involved the most electricity consumption and labour, making these the main contributors to carbon emissions. Conversely, the dyeing phase was characterized by its extensive chemical usage, this study does not consider the impact of garment colour on CFP.

2.3. Data collection

The research was underpinned by survey data and the China Statistical Yearbook. Cotton cultivation figures were sourced from the China Statistical Yearbook. Survey data collected in 2022 comprised 60 valid responses, primarily from factories in Xinjiang and Henan Provinces,

focusing on production aspects. Acknowledging the challenge of gathering commercially sensitive information, ginning mills in Xinjiang, several major textile manufacturers, and childrenswear industrial parks known for their industrial concentrations were specifically targeted in this study. The aim of this approach was to collect comprehensive data across all stages of industrial production: ginning, spinning, knitting, dyeing, and final garment making-up, as detailed in Table A.2. The production and operational management of large textile enterprises in China were considered somewhat representative, and the data collected from these companies were considered representative. This study focuses on providing specific information about manufacturers or technologies rather than generalizing data to illustrate overall impacts and key areas.

2.4. Data calculation

Within the realm of industrial production, the encompassing CFP comprises emissions derived from production processes, administrative operations, and auxiliary segments. Specifically, emissions attributed to management activities are sourced from administrative and public services unrelated to direct production. Conversely, auxiliary emissions are associated with ancillary production activities, such as internal transportation and facility depreciation (Wang et al., 2015). In this investigation, both the direct production CFP and the ancillary CFP within industrial production settings were examined.

The CFP was derived utilizing the emission factor methodology delineated by the IPCC (IPCC, 2019), a technique that has gained widespread acceptance (CarbonTrust, 2014). This method involves quantifying GHG emissions in terms of carbon dioxide equivalents, which are correlated with specific activity units. In agreement with the carbon emission inventory, a

comprehensive model representing the entire cotton industry chain was developed. This model could estimate carbon emissions for various projects by multiplying the input energy consumption by the respective emission factors. The specific inputs and emission factors for both the cultivation phase and the subsequent processing stages are detailed in Table A.2.

2.5. Scenario assumptions

According to literature reviews and practical observations, four distinct scenarios were proposed in this study. Due to the considerable variations in emissions across factories of different sizes, a scenario based on scale differences was established. Organic cotton, defined as cotton grown without synthetic agricultural chemicals such as fertilizers, pesticides, and genetically modified organisms (Günther et al., 2017; Delate et al., 2021), could reduce emissions if utilized in garment production, prompting the substitution of traditional cotton cultivation with organic cotton (Gonzalez et al., 2023). With the gradual replacement of traditional coal-fired power generation with alternative energy sources, such as hydroelectric, wind, and solar power—which effectively reduce the CFP—the scenario for wind power generation was modelled by substituting the carbon emission factor of hydroelectric power for the general electricity consumption factor. Additionally, recycling textiles, which could bypass the cotton cultivation or dyeing stages, led to the establishment of a textile recycling scenario.

3. Results

3.1. Carbon footprints of various types of garments with different measurement units

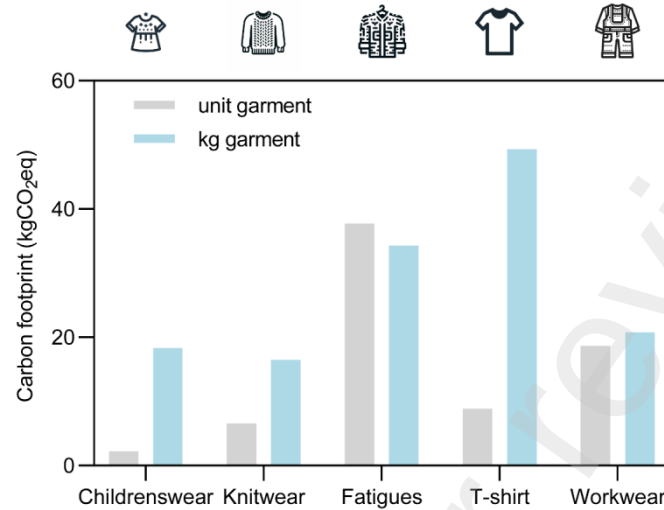
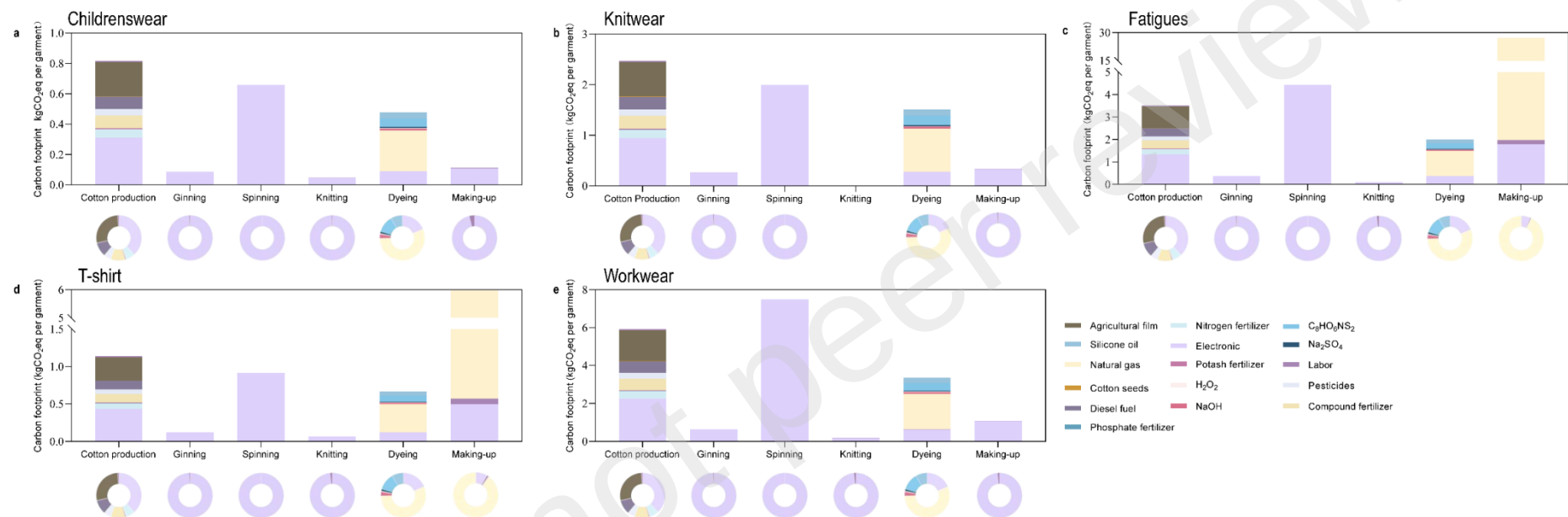


Fig. 2. CFP comparison across clothing types, illustrated with grey bars for the footprint of individual items and blue bars for the footprint per kilogram.

The CFPs associated with various types of clothing demonstrate significant variations (Fig. 2). For instance, the CFPs for cotton childrenswear (0.12 kg), cotton knitwear (0.40 kg), nylon/cotton fatigues (1.10 kg), T-shirts (0.18 kg), and cotton workwear (0.90 kg) are 2.20, 6.58, 37.70, 8.88 and 18.68 kgCO₂eq, respectively. When assessing the CFP per unit of clothing, the fatigues exhibit the greatest footprint, surpassing those of childrenswear, knitwear, T-shirts, and workwear by 94.16%, 82.55%, 76.44%, and 50.46%, respectively. Conversely, when the functional unit is standardized by weight, the CFPs are 18.35 kgCO₂eq for childrenswear, 16.45 kgCO₂eq for knitwear, 34.28 kgCO₂eq for fatigues, 49.34 kgCO₂eq for T-shirts, and 20.75 kgCO₂eq for workwear. In this scenario, T-shirts have the largest CFP per unit weight, exceeding those of childrenswear, knitwear, fatigues, and workwear by 62.82%, 66.66%, 30.53%, and 57.94%, respectively.

204 3.2. Carbon emission levels and compositions at different stages of garment production

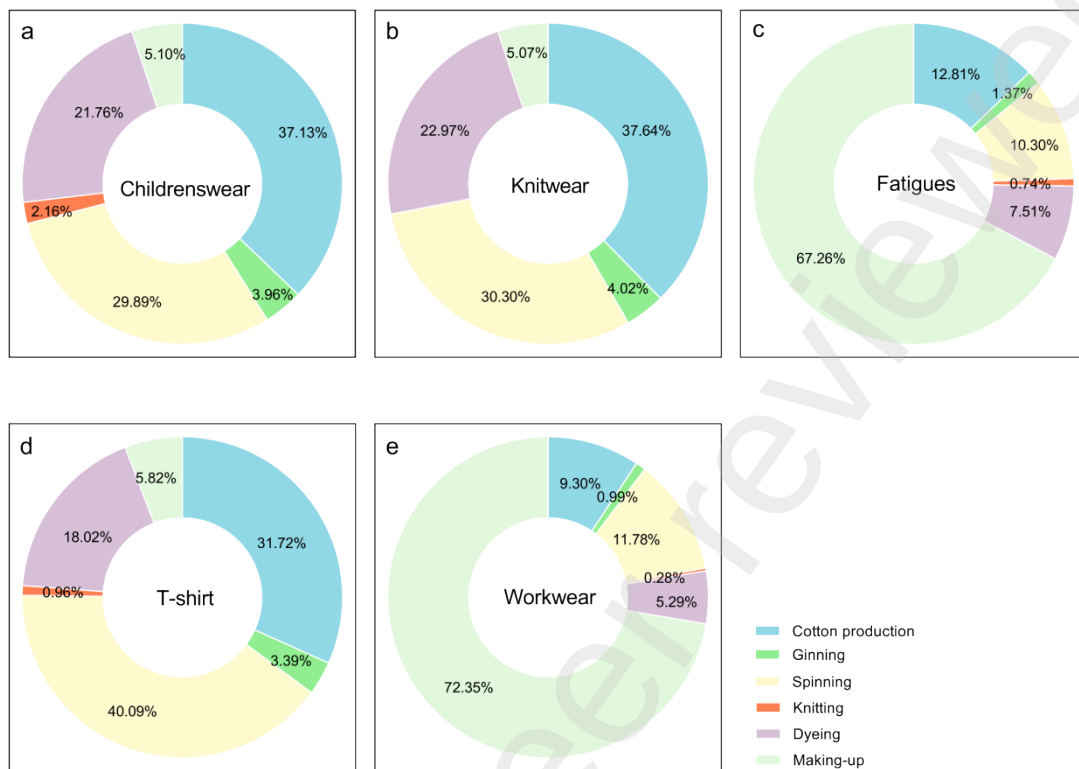


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206 **Fig. 3.** Carbon emission composition at each stage of five types of garment production. Fig. 3a to e represent the CFP of producing childrenswear, knitwear,
207 fatigues, T-shirt and workwear respectively (1 piece of garment).

CFP analysis reveals that the cotton production stage is the most significant contributor to the CFP of childrenswear production, representing 37.13% of the total industrial chain with 0.82 kgCO₂eq (Fig. 3a). Electricity consumption (0.31 kgCO₂eq, accounting for 38.15%) and the utilization of agricultural films (0.23 kgCO₂eq, representing 27.57%) are the primary factors affecting the cultivation phase. The spinning phase has the second largest impact at 0.66 kgCO₂eq, accounting for 29.89% of the total CFP across the industry chain (Fig. 4a), where electricity consumption constitutes more than 99% of its composition. Similar high electricity dependencies are observed in the ginning, knitting, and garment making-up stages, each exceeding 99% of the electricity consumption in their emissions profiles. The dyeing stage, accounting for 21.76% of the total CFP (0.48 kgCO₂eq), exhibits a distinct emission profile where steam usage is the predominant factor, contributing 56% (0.27 kgCO₂eq) to its CFP.

The cotton production process of knitwear has the highest carbon emissions among all the stages, contributing 37.64% (2.48 kgCO₂eq) to the overall emissions (Fig. 3b). Notably, the carbon emissions from electricity use (0.95 kgCO₂eq, 38.15%) and agricultural film application (0.68 kgCO₂eq, 27.57%) are significantly greater than those from other inputs in the cotton production stage. The spinning stage follows, accounting for 22.97% of the total CFP with 1.99 kgCO₂eq (Fig. 4b). Therefore, electricity consumption is the primary contributor to the CFP, comprising more than 99%, while the CFP from labour in this stage is less than 1%. The ginning and garment manufacturing stages exhibit relatively low emissions, at 0.26 kgCO₂eq (4.02%) and 0.33 kgCO₂eq (5.07%), respectively, mirroring the spinning stage in their emissions composition. The dyeing stage, marked by substantial steam consumption, results in carbon emissions of 0.85 kgCO₂eq. Importantly, the knitwear production process bypasses knitting,

230 leading to a minimal CFP at this stage.



231

232 **Fig. 4.** The levels of carbon emissions at each stage of five types of garment production. Fig.
 233 4a to e represent the percentages of each stage in the production of childrenswear, knitwear,
 234 fatigues, T-shirt and workwear respectively.

235 In the production of fatigues, the garment making-up stage has the largest CFP, totalling
 236 27.28 kgCO₂eq and accounting for 72.35% of the CFP of the entire production process (Fig.
 237 3c). The use of steam contributes the most to carbon emissions, amounting to 25.30 kgCO₂eq
 238 and accounting for 92.75% of the CFP at this stage. The spinning and cotton production stages
 239 contribute 11.78% and 9.30%, respectively, to the total CFP, with emissions of 4.44 kgCO₂eq
 240 and 3.51 kgCO₂eq, respectively (Fig. 4c). In the spinning phase, power consumption is the
 241 primary source of the CFP, exceeding 99%. Conversely, the knitting stage has the smallest
 242 carbon impact at only 0.28% (0.11 kgCO₂eq), where similarly, electricity usage is the primary

emission source.

According to the life cycle analysis of T-shirts and fatigues, the CFP distribution and components across stages are similar, with the making-up stage showing the greatest contribution at 5.97 kgCO₂eq, accounting for 72.35% of the total emissions (Fig. 3d). In this phase, the use of steam is the primary source of carbon emissions, amounting to 5.40 kgCO₂eq, which constitutes 92.75% of the CFP generated. The cultivation stage is the second largest contributor to the CFP in T-shirt production, with electricity use (0.43 kgCO₂eq, 38.15%) and agricultural film application (0.31 kgCO₂eq, 27.57%) as the primary factors. The ginning, spinning, and knitting stages are ranked fifth, third, and fourth in terms of their carbon shares (Fig. 4d), respectively, with electricity consumption being the predominant factor in these stages, accounting for more than 99% of their emissions.

When producing a piece of workwear, the spinning stage is the phase that contributes the most to total carbon emissions, amounting to 7.49 kgCO₂eq, which constitutes 40.09% of the total CFP across the supply chain (Fig. 3e). This factor differs from the production of other types of clothing. The cotton production stage is the second largest contributor, accounting for 31.72% of the total emissions (Fig. 4e), with a value of 5.92 kgCO₂eq. Within this stage, electricity consumption (2.26 kgCO₂eq, 38.15%) and agricultural film use (1.63 kgCO₂eq, 27.57%) are key factors. The dyeing phase ranks third in terms of carbon emissions at 3.37 kgCO₂eq, accounting for 18.02% of the total CFP across the supply chain. The use of steam is the primary contributor to the CFP at this stage, accounting for 56% of the total CFP (1.88 kgCO₂eq). For the spinning, making-up, ginning, and knitting stages, electricity consumption dominates the CFP, accounting for 98.35% of the CFP of the knitting stage of more than 99%

of the CFPs of the other stages.

3.3. Scenario analysis

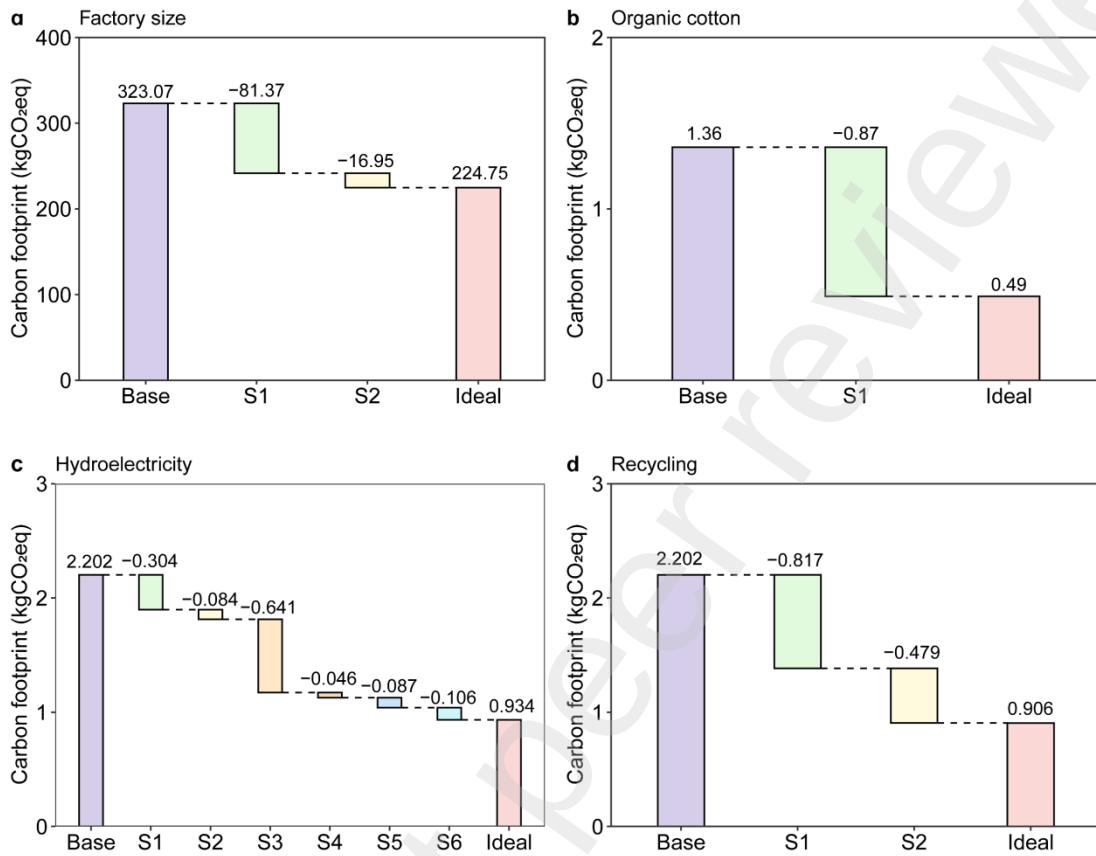


Fig. 5. Emission reduction benefits of different scenarios: (a) impact of the ginning plant size on the CFP size; S1 and S2 represent ginning plant capacities greater than 10,000 tonnes of lint and less than 7,000 tonnes of lint, respectively; (b) impact of organic and conventional cotton cultivation on the CFP; (c) analysis of electricity substitution scenarios. S1 to S6 represent the reduction in CFP at each stage of electricity substitution, and (d) shows a comparison of the reduction in emissions from garment recycling. S1 and S2 represent the reduction in the cotton production stage and the dyeing stage, respectively.

Analysis of 20 ginning mills reveals an inverse relationship between mill capacity and environmental impact, with both the largest (10,000 tonnes of lint) and smallest (4,800 tonnes

of lint) capacities outperforming medium-scale production in terms of environmental sustainability (Fig. 5a). Specifically, mills exceeding 10,000 tonnes in capacity have 25.19% smaller CFPs, whereas those with capacities below 7,000 tonnes exhibit reductions of 30.43% compared to the average footprint of the analysed gins. According to the study by Gonzalez et al., (2023), calculations indicate that transitioning to organic cotton cultivation can reduce the CFP of a single piece of childrenswear during the cotton production phase by 64.2% (Fig. 5b). This reduction chiefly arises due to decreased fertilizer application and nitrous oxide emissions from agricultural fields. By substituting the coal energy source with hydropower, the CFP of cotton childrenswear can be reduced by 57.58% throughout its life cycle. The impacts are distributed across the cotton production, ginning, spinning, knitting, dyeing, and making-up stages, with reductions of 13.81%, 3.81%, 29.11%, 2.09%, 3.95%, and 4.81%, respectively (Fig. 5c). Omitting the cultivation phase can reduce the overall carbon emissions of a cotton childrenswear life cycle by 37.10%, while removing both the cultivation and dyeing stages results in a 58.85% reduction, with the dyeing stage alone contributing to a 21.75% decrease in overall emissions (Fig. 5d).

4. Discussion

4.1. From seed to garment: Carbon footprints of Chinese cotton textiles

The cotton textile industry chain is lengthy and complex, with its production, processing, and usage stages contributing to varying degrees of environmental impacts. Comprehensive analyses of literature and databases over the past two decades have indicated the presence of significant regional differences in the environmental impacts associated with cotton cultivation, manufacturing, and consumer behaviour (Zhang et al., 2023). Depending on the type of impact

and the country, cotton cultivation, manufacturing, and usage can have major effects on the environment. This influence underscores the importance of assessing the environmental impacts of cotton textile production and processing in specific regions. As a country that greatly contributes to the industries of cotton textile production, consumption, and trade, conducting CFP and environmental footprint analyses for different cotton textile products in China is highly significant. Using a single garment as a functional unit, the CFPs of different types of clothing vary substantially (Table 1), with a range from 2.20 to 37.70 kg CO₂eq for five different categories being reported in this study. This aligns with the findings of Wagner et al. (2023), where a comprehensive synthesis of the literature reveals significant differences in the CFP of various clothing types. Wang et al. (2015) reported that the average life cycle CFP of cotton shirts produced in China is 8.77 kgCO₂eq, this value falls within the range of the results of this study. This variation is attributable to the different weights, processing techniques, and processes of different clothes.

Table 1

Carbon footprints of different garments at different system boundaries

Function unit (1 piece)	System boundary	Carbon footprint (kg CO ₂ eq)	Reference resource
Childrenswear	Seed-to-garment	2.20	This study
Knitwear	Seed-to-garment	6.58	This study
Fatigues	Seed-to-garment	37.70	This study
T-shirt	Seed-to-garment	8.88	This study
Workwear	Seed-to-garment	18.68	This study
Polo shirt	Cradle-to-grave	13.90	Wagner et al., 2023
Doctor's coat	Cradle-to-grave	33.10	Wagner et al., 2023
Scrub trousers	Cradle-to-grave	15.50	Wagner et al., 2023
Shirt	Cradle-to-use	8.77	Wang et al., 2015
Trouser jeans	Cradle-to-garment	7.86	Morita et al., 2020
Jeans	Cradle-to-grave	90.37	Luo et al., 2022

Although the CFPs of individual garments can quantify the total emissions from their production, assessing the CFP per kilogram of a garment enables the comparison of the

environmental efficiencies across various materials and manufacturing techniques. The cotton production phase is pivotal in the cotton life cycle, where the quality of seed cotton fundamentally influences the quality of the resultant textiles. In this study, it is revealed that the CFP associated with the production of 1 kg of seed cotton is 4.92 kgCO₂eq, diverging from findings of Huang et al. (2022) . This discrepancy arises from our use of the latest statistical data, which focuses specifically on Xinjiang Province. The CFP during the cultivation phase is most significantly affected by electricity consumption, the use of agricultural film, and the application of fertilizers. The environmental impact during the cultivation stage can be reduced by decreasing the use of nitrogen fertilizers and improving agricultural practices (Bansal et al., 2022). Concurrently, Huang et al. (2022) reported that the use of agricultural film for cotton cultivation in Xinjiang Province is greater than that in other provinces. Considering the gradual shift of cotton cultivation to Xinjiang Province, reducing the CFP of this phase of cotton cultivation is a technical challenge that must be addressed.

The CFP of the spinning stage varies for each type of garment, with workwear having the highest footprint at 7.49 kg CO₂eq, followed by fatigues at 4.44 kg CO₂eq. The variability arises from the detailed segmentation of production processes beginning with spinning, during which the yarn count fundamentally determines the quality of the final textile. An increased yarn count signifies enhanced clothing quality. The specific subprocesses vary according to factors such as the required yarn fineness and fabric type (e.g., twill or plain), which influences the selection of materials and electricity consumption.

Within the entire life cycle, the CFP attributed to the knitting stage consistently represents the smallest percentage, at 2.16% for childrenswear, while for the other four garment categories,

it remains below 1%. In contrast to this trend, Wang et al. (2015) identified the knitting stage as the second largest contributor to the life cycle CFP of cotton shirts manufactured in China, differing from the results presented in this article. This contribution arises because the scale and geographic location of factories greatly affect the CFP of apparel.

The CFPs during the making-up stages of both T-shirts and fatigues contribute the most to the total CFPs of the whole production process, with percentages of 67.26% and 72.35%, respectively. The results of this study indicate that natural gas is a major contributor to the making-up stage, and in order to generate steam, a large amount of energy is consumed, resulting in more greenhouse gas emissions. Similarly, echoing our results, Luo et al. (2022) noted that this stage contributes the most to the overall CFP when steam is the primary emission source. Additionally, the method of steam transportation via pipelines in this facility incurs substantial energy losses. However, this scenario, which is specific to a factory leveraging weather-generated steam, may not universally represent the garment manufacturing sector. In contrast, the making-up stage of childrenswear apparel exhibits a markedly reduced CFP at 5.10%, which is attributed to the strategic location of these factories within highly aggregated apparel industrial parks, where energy management is centralized. To enhance environmental sustainability, particularly in small-to medium-sized and park-based factories, the adoption of electric boilers for steam generation is recommended, signifying a potential shift towards increasingly energy-efficient garment production processes.

Economic considerations play a pivotal role in shaping the environmental footprints of cotton textiles across their production chain. In the initial processing stages, ginning mills emphasize the yield ratio of cottonseed to cotton and the quality of seed cotton, which are

crucial for operational efficiency. Fluctuations in market prices and the involvement of capital investors further exacerbate this challenge. Our findings indicate that in response to market downturns, ginning mills adopt a cautious approach by scaling down production rather than ceasing operations, aiming to minimize operational costs. This scaling reduces the amount of finished lint processed per unit time, but the relative energy consumption does not decrease annually; thus, the carbon footprint per unit of output increases. Similar circumstances are encountered in factories across other production stages, where the influences of economic factors on the CFP are equally significant and cannot be overlooked (Christis et al., 2019).

4.2. From seed-to-garment: Potential trends in reducing the carbon footprint of Chinese cotton textiles

This study indicated that when ginning mills operate at relatively large or small scales, their CFPs decrease, with reductions of 25.19% and 30.43%, respectively. Large-scale production can mitigate the overall environmental impact by increasing the efficiency of energy utilization and reducing the allocation of fixed costs per unit of product. Large factories may use advanced technologies to efficiently use energy and reduce energy consumption and carbon emissions per unit of product (Porter and Van der Linde, 1995). Conversely, smaller ginning factories may exhibit a reduced CFP, which can be attributed to their diminished production scales. These manufacturers often depend on manual or semiautomated processes that, despite being less efficient, potentially minimize energy consumption by eschewing the operation of large-scale, energy-intensive machinery. Moreover, the impact of scale on the carbon footprint is related to energy management and efficiency improvement measures during the production process. Large entities might allocate additional resources towards the adoption of renewable

energy sources and advanced, efficient apparatuses. In contrast, small establishments might predominantly utilize conventional methods and energy solutions.

In term of present results, compared with conventional cotton, organic cotton offers greater environmental benefits. Gonzalez et al., (2023) reported that the CFP of organic cotton is 64.20% smaller than that of conventional cotton. This difference is mainly due to the lack of inputs of chemical fertilizers and pesticides in the organic cotton production stage, reducing insect pests and improving soil quality through crop rotation (Delate et al., 2021), which subsequently decreases indirect field emissions. However, given the high quality of organic cotton and the expenses involved in its intensive, large-scale farming, the global market share of organic cotton remains considerably low. This study shows that the planting phase accounts for a substantial proportion of carbon emissions from cotton textiles; thus, intensifying the promotion of organic cotton may serve as a key approach for reducing emissions from cotton textiles in China. This shift reduces the reliance on synthetic agrochemicals and enhances ecosystem carbon sequestration, benefiting soil health and biodiversity (Delate et al., 2021; He, 2023). Nonetheless, the generally lower yields of organic matter compared to those of conventional cotton are challenging, requiring careful analysis in its application.

In tracking the CFP of the clothing industry, we found that electricity constitutes a significant proportion of emissions throughout the entire cotton industry chain. This finding aligns with Luo et al. (2022), who reported that electricity, steam, diesel, and other energy sources collectively account for more than 90% of the emissions in each garment processing stage. Transitioning from conventional power generation to hydroelectricity can dramatically reduce the CFP of childrenswear by 57.58% (Fig. 8c). Karunathilake et al. (2018) explored the

beneficial effects on emission reduction achieved by integrating renewable energy into a power mix. Additionally, researchers have discovered that using wood as a heat source results in the lowest CFP for jeans (Morita et al., 2020). This finding mirrors our results, which indicate that replacing energy sources like natural gas and coal with alternatives can significantly reduce the CFP of clothing. Transitioning the source of electricity from coal to clean alternatives, such as hydroelectricity, nuclear power, and renewable energies, can reduce the GHG emission (Huang et al., 2017). Then, the carbon emissions of the entire textile industry chain decrease.

Current study results suggested that recycling textiles can eliminate the planting and dyeing stages in the entire industrial chain, reducing the CFP of childrenswear by 58.85%. Previous study have demonstrated that altering the methods of using and processing clothing can effectively reduce the CFP (Overcash, 2012). Compared with direct disposal or incineration, recycling and reusing discarded garments are efficient strategies for reducing the environmental impact, resulting in the least environmental burden (Xie et al., 2021). This finding highlights the importance of improving the efficiency of recycling waste clothing, which plays a key role in promoting the sustainable development of the clothing industry (Mora-Sojo et al., 2023; Wiedemann et al., 2023). By recycling and reusing textiles, we can reduce the demand for raw materials and associated carbon emissions and foster a transition of the industry towards an increasingly sustainable production and consumption model (Sapkal, 2018).

4.3. Limitations and perspectives

The diversity of data sources increases the complexity of the research. The use of statistical yearbooks and field surveys may introduce bias. The statistical yearbook data are official, and it is impossible to evaluate the reliability of the data. Additional information must be relied

upon for in-depth research. When conducting CFP assessments, a large amount of primary data must be collected; these data are usually obtained through on-site surveys. Data from research are often gathered using standardized data sheets from factories or direct measurements from production equipment. The veracity of these data can be enhanced by cross-referencing the aggregated data from individual subprocesses with overall organizational material and energy consumption statistics.

The primary challenge in assessing CFPs across clothing life cycles in this study arises from the vast diversity of clothing types. To enhance the rigor of the assessment, our framework aligns with established methodologies from the International Standards Organization (ISO) and the IPCC, aiming for thorough coverage of key processes in the life cycles of five distinct types of clothing. Consequently, our methodology exhibits enhanced thoroughness. In the factories we surveyed, there is a tendency to specialize in the production of specific types of garments and products. For instance, the garment factory producing fatigues has, over the past five years, dedicated 90% of its total output to the manufacturing of these uniforms. The characteristics of this supply chain present a compelling case study, offering insights into the mechanisms by which the production of a single type of product can impact the CFP. An analysis of the environmental impact of jeans produced in Brazil highlights the impact of a single type of garment on the energy consumption (Morita et al., 2020), without considering various clothing categories. Our work fills this gap by providing a systematic reference for comparing CFPs across different clothing life cycles.

Approximately 80% of textile enterprises in China are small or medium-sized businesses, and they are geographically dispersed (Chi, 2011). This factor contributes to the suboptimal

environmental performance of the apparel sector (Kostka et al., 2013). Therefore, reconfiguring the entire industry chain model for clothing has significant environmental advantages. For future research, it is advisable to build upon this study by disaggregating textile categories. This approach can broaden the selection of textile types for analysis and enable precise quantification of carbon emissions for each textile type at every stage of its life cycle. The carbon emissions at each stage of the life cycle for various textiles are quantified, thereby enhancing the specificity and detail of the assessment results.

5. Conclusions

This research quantifies the CFP of five clothing types produced in China, revealing significant variations. The cotton production, spinning, and making-up stages were identified as the highest emitters, with CFPs ranging from 2.20 kgCO₂eq for childrenswear to 37.70 kgCO₂eq for fatigues. Notably, the making-up stage is the main contributor for fatigues, while cotton production is for childrenswear. Industrial electricity consumption was the major pollution source, and scenario-based simulations showed that switching from coal to hydroelectric power significantly reduces emissions. This study provides essential data for stakeholders in the cotton industry chain to develop targeted strategies for improving energy efficiency and reducing emissions. It also underscores the importance of comprehensive strategies across the apparel supply chain, including design improvements, increased garment longevity, strict quality control, and circular design principles. Enhancing supply chain transparency and refining textile recycling methods are crucial for advancing sustainability in the apparel industry.

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